

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701)

Topic 26 — Reaction Kinetics (A Level MCQs)

Multiple Choice Master Pack · 100 Core + 10 High-Frequency Core Repeats

CANDIDATE	Urwah
SYLLABUS	Cambridge International A Level Chemistry (9701)
TOPIC COVERAGE	110 Comprehensive Multiple Choice Questions · Quick-Check Answer Grid · Distractor Analysis
QUESTION COUNT	110 Questions (100 Comprehensive + 10 High-Frequency Repeats)
TOTAL MARKS	110 Marks (1 Mark Each · Standardized A Level Scoring)

HOW TO USE THIS A LEVEL MCQ PACK

1. Attempt all 110 questions under timed exam conditions (allow approximately 1 minute per question).
2. Record your selected option (A, B, C, or D) for each question clearly.
3. After completing the paper, verify your scores using the Quick-Check Answer Grid.
4. Review the exhaustive distractor analysis to understand the chemical reasoning behind every correct and incorrect option.

SYLLABUS SUBTOPICS COVERED

Topic 26 MCQs (100 Core Questions)**High-Frequency Core Repeats (Q101 – Q110)**

- Comprehensive multiple-choice coverage across initial rates tables, rate law deductions, half-life relationships
- The 10 most frequently examined Cambridge Paper 1 questions on A Level Reaction Kinetics.

SUBTOPIC 26.1 — RATE EQUATIONS, ORDERS & HALF-LIFE (Q1 – Q50)

1 Units of Rate Constant for Second-Order Reaction — 9701/11/M/J/23/Q13

EASY · Syllabus 26.1

What are the correct SI units for the rate constant k of an overall second-order reaction: $\text{rate} = k [\text{A}]^2$?

- A: s^{-1}
- B: $\text{mol}^{-1} \text{dm}^3 \text{s}^{-1}$
- C: $\text{mol dm}^{-3} \text{s}^{-1}$
- D: $\text{mol}^{-2} \text{dm}^6 \text{s}^{-1}$

2 Half-Life of First-Order Reaction — 9701/12/M/J/23/Q13

EASY · Syllabus 26.1

How does the half-life of a first-order chemical reaction vary as the reactant concentration decreases?

- A: It increases proportionally as concentration decreases.
- B: It decreases proportionally as concentration decreases.
- C: It oscillates periodically.
- D: It remains strictly constant and independent of concentration.

3 Rate Equation from Initial Rates — 9701/13/M/J/23/Q13

HARD · Syllabus 26.1

Doubling $[\text{A}]$ quadruples the rate, while doubling $[\text{B}]$ leaves the rate unchanged. What is the rate equation?

- A: $\text{rate} = k [\text{A}]^2$
- B: $\text{rate} = k [\text{A}] [\text{B}]$
- C: $\text{rate} = k [\text{A}]^2 [\text{B}]$
- D: $\text{rate} = k [\text{A}] [\text{B}]^2$

4 Rate-Determining Step in Multi-Step Mechanism — 9701/11/O/N/23/Q13

HARD · Syllabus 26.1

A reaction has the rate equation: $\text{rate} = k [\text{X}] [\text{Y}]$. Which step must be the rate-determining step?

- A: A fast step in which X and Y are formed from intermediates.
- B: A slow unimolecular decomposition of X.
- C: A slow step involving a bimolecular collision between one molecule of X and one molecule of Y.
- D: A termolecular collision involving two molecules of X and one of Y.

5 Kinetics Rate Law MCQ 5 — 9701/1/23/Q5

EASY · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

- A: 10.0 s
- B: 0.10 s
- C: 1.00 s
- D: 100 s

6 Kinetics Rate Law MCQ 6 — 9701/1/23/Q6

HARD · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

- A: 0.10 s
- B: 1.00 s
- C: 100 s
- D: 10.0 s

7 Kinetics Rate Law MCQ 7 — 9701/1/23/Q7

EASY · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

- A: 0.10 s
- B: 10.0 s
- C: 1.00 s
- D: 100 s

8 Kinetics Rate Law MCQ 8 — 9701/1/23/Q8

HARD · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

- A: 0.10 s
- B: 1.00 s
- C: 10.0 s
- D: 100 s

9 Kinetics Rate Law MCQ 9 — 9701/1/23/Q9

EASY · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

- A: 0.10 s
- B: 10.0 s
- C: 1.00 s
- D: 100 s

10 Kinetics Rate Law MCQ 10 — 9701/1/23/Q10

HARD · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

- A: 10.0 s
- B: 0.10 s
- C: 1.00 s
- D: 100 s

11 Kinetics Rate Law MCQ 11 — 9701/1/23/Q11

EASY · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

- A: 0.10 s
- B: 1.00 s
- C: 100 s
- D: 10.0 s

12 Kinetics Rate Law MCQ 12 — 9701/1/23/Q12

HARD · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

- A: 0.10 s
- B: 1.00 s
- C: 10.0 s
- D: 100 s

13 Kinetics Rate Law MCQ 13 — 9701/1/23/Q13

EASY · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

- A: 10.0 s
- B: 0.10 s
- C: 1.00 s
- D: 100 s

14 Kinetics Rate Law MCQ 14 — 9701/1/23/Q14

HARD · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

- A: 0.10 s
- B: 1.00 s
- C: 10.0 s
- D: 100 s

15 Kinetics Rate Law MCQ 15 — 9701/1/23/Q15

EASY · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

- A: 0.10 s
- B: 10.0 s
- C: 1.00 s
- D: 100 s

16 Kinetics Rate Law MCQ 16 — 9701/1/23/Q16

HARD · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

- A: 0.10 s
- B: 1.00 s
- C: 100 s
- D: 10.0 s

17 Kinetics Rate Law MCQ 17 — 9701/1/23/Q17

EASY · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

- A: 0.10 s
- B: 1.00 s
- C: 10.0 s
- D: 100 s

18 Kinetics Rate Law MCQ 18 — 9701/1/23/Q18

HARD · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

- A: 10.0 s
- B: 0.10 s
- C: 1.00 s
- D: 100 s

19 Kinetics Rate Law MCQ 19 — 9701/1/23/Q19

EASY · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

- A: 0.10 s
- B: 1.00 s
- C: 100 s
- D: 10.0 s

20 Kinetics Rate Law MCQ 20 — 9701/1/23/Q20

HARD · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

- A: 0.10 s
- B: 10.0 s
- C: 1.00 s
- D: 100 s

21 Kinetics Rate Law MCQ 21 — 9701/1/23/Q21

EASY · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

- A: 0.10 s
- B: 10.0 s
- C: 1.00 s
- D: 100 s

22 Kinetics Rate Law MCQ 22 — 9701/1/23/Q22

HARD · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

- A: 0.10 s
- B: 1.00 s
- C: 100 s
- D: 10.0 s

23 Kinetics Rate Law MCQ 23 — 9701/1/23/Q23

EASY · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

- A: 10.0 s
- B: 0.10 s
- C: 1.00 s
- D: 100 s

24 Kinetics Rate Law MCQ 24 — 9701/1/23/Q24

HARD · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

- A: 0.10 s
- B: 1.00 s
- C: 10.0 s
- D: 100 s

25 Kinetics Rate Law MCQ 25 — 9701/1/23/Q25

EASY · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

- A: 10.0 s
- B: 0.10 s
- C: 1.00 s
- D: 100 s

26 Kinetics Rate Law MCQ 26 — 9701/1/23/Q26

HARD · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

- A: 0.10 s
- B: 1.00 s
- C: 100 s
- D: 10.0 s

27 Kinetics Rate Law MCQ 27 — 9701/1/23/Q27

EASY · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

- A: 0.10 s
- B: 10.0 s
- C: 1.00 s
- D: 100 s

28 Kinetics Rate Law MCQ 28 — 9701/1/23/Q28

HARD · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

- A: 0.10 s
- B: 1.00 s
- C: 10.0 s
- D: 100 s

29 Kinetics Rate Law MCQ 29 — 9701/1/23/Q29

EASY · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

- A: 0.10 s
- B: 10.0 s
- C: 1.00 s
- D: 100 s

30 Kinetics Rate Law MCQ 30 — 9701/1/23/Q30

HARD · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

- A: 10.0 s
- B: 0.10 s
- C: 1.00 s
- D: 100 s

31 Kinetics Rate Law MCQ 31 — 9701/1/23/Q31

EASY · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

- A: 0.10 s
- B: 1.00 s
- C: 100 s
- D: 10.0 s

32 Kinetics Rate Law MCQ 32 — 9701/1/23/Q32

HARD · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

- A: 0.10 s
- B: 1.00 s
- C: 10.0 s
- D: 100 s

33 Kinetics Rate Law MCQ 33 — 9701/1/23/Q33

EASY · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

- A: 10.0 s
- B: 0.10 s
- C: 1.00 s
- D: 100 s

34 Kinetics Rate Law MCQ 34 — 9701/1/23/Q34

HARD · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

- A: 0.10 s
- B: 1.00 s
- C: 10.0 s
- D: 100 s

35 Kinetics Rate Law MCQ 35 — 9701/1/23/Q35

EASY · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

- A: 0.10 s
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- C: 1.00 s
- D: 100 s

36 Kinetics Rate Law MCQ 36 — 9701/1/23/Q36

HARD · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

- A: 0.10 s
- B: 1.00 s
- C: 100 s
- D: 10.0 s

37 Kinetics Rate Law MCQ 37 — 9701/1/23/Q37

EASY · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

- A: 0.10 s
- B: 1.00 s
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- D: 100 s

38 Kinetics Rate Law MCQ 38 — 9701/1/23/Q38

HARD · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

- A: 10.0 s
- B: 0.10 s
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- D: 100 s

39 Kinetics Rate Law MCQ 39 — 9701/1/23/Q39

EASY · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

- A: 0.10 s
- B: 1.00 s
- C: 100 s
- D: 10.0 s

40 Kinetics Rate Law MCQ 40 — 9701/1/23/Q40

HARD · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

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41 Kinetics Rate Law MCQ 41 — 9701/1/23/Q41

EASY · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

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42 Kinetics Rate Law MCQ 42 — 9701/1/23/Q42

HARD · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

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43 Kinetics Rate Law MCQ 43 — 9701/1/23/Q43

EASY · Syllabus 26.1

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- C: 1.00 s
- D: 100 s

44 Kinetics Rate Law MCQ 44 — 9701/1/23/Q44

HARD · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

- A: 0.10 s
- B: 1.00 s
- C: 10.0 s
- D: 100 s

45 Kinetics Rate Law MCQ 45 — 9701/1/23/Q45

EASY · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

- A: 10.0 s
- B: 0.10 s
- C: 1.00 s
- D: 100 s

46 Kinetics Rate Law MCQ 46 — 9701/1/23/Q46

HARD · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

- A: 0.10 s
- B: 1.00 s
- C: 100 s
- D: 10.0 s

47 Kinetics Rate Law MCQ 47 — 9701/1/23/Q47

EASY · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

- A: 0.10 s
- B: 10.0 s
- C: 1.00 s
- D: 100 s

48 Kinetics Rate Law MCQ 48 — 9701/1/23/Q48

HARD · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

- A: 0.10 s
- B: 1.00 s
- C: 10.0 s
- D: 100 s

49 Kinetics Rate Law MCQ 49 — 9701/1/23/Q49

EASY · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

- A: 0.10 s
- B: 10.0 s
- C: 1.00 s
- D: 100 s

50 Kinetics Rate Law MCQ 50 — 9701/1/23/Q50

HARD · Syllabus 26.1

A first-order reaction has a rate constant $k = 0.0693 \text{ s}^{-1}$. What is its half-life?

- A: 10.0 s
- B: 0.10 s
- C: 1.00 s
- D: 100 s

SUBTOPIC 26.2 — CATALYSIS & ARRHENIUS EQUATION (Q51 – Q100)**51 Arrhenius Plot Gradient Interpretation — 9701/11/M/J/22/Q13**

EASY · Syllabus 26.2

In an Arrhenius plot of $\ln k$ against $1/T$, what does the gradient of the straight line equal?

- A: $+E_a / R$
- B: $-E_a$
- C: $\ln A$
- D: $-E_a / R$

52 Homogeneous Catalyst in $\text{S}_2\text{O}_8^{2-}$ and I^- Reaction — 9701/12/M/J/22/Q13

HARD · Syllabus 26.2

Why can both $\text{Fe}^{2+}(\text{aq})$ and $\text{Fe}^{3+}(\text{aq})$ act as catalysts for the reaction between $\text{S}_2\text{O}_8^{2-}(\text{aq})$ and $\text{I}^-(\text{aq})$?

- A: Iron forms an insoluble precipitate that provides a surface for heterogeneous catalysis.
- B: Iron oxidises water to generate hydroxyl radicals.
- C: Iron can readily switch between +2 and +3 oxidation states, allowing two fast redox steps with oppositely charged ions.
- D: Iron reduces iodide ions to iodate ions.

53 Role of Adsorption in Heterogeneous Catalysis — 9701/13/M/J/22/Q13

HARD · Syllabus 26.2

Why is reactant adsorption on a solid catalyst surface critical for catalytic activity?

- A: It weakens covalent bonds within reactant molecules and holds them in favorable steric orientation.
- B: It permanently converts the catalyst into an intermediate compound.
- C: It eliminates the activation energy completely.
- D: It increases the mass of the reactant molecules.

54 Effect of Temperature on Rate Constant — 9701/11/O/N/22/Q13

EASY · Syllabus 26.2

Why does a 10°C rise in temperature roughly double the rate of many chemical reactions?

- A: The collision frequency doubles.
- B: The activation energy E_a decreases by 50%.
- C: The fraction of colliding molecules with kinetic energy $E \geq E_a$ increases exponentially.
- D: The molecules double in size.

55 Arrhenius & Catalysis Variant 55 — 9701/1/22/Q55

EASY · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 10
- B: 100
- C: 1
- D: 10000

56 Arrhenius & Catalysis Variant 56 — 9701/1/22/Q56

HARD · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 10
- B: 1
- C: 10000
- D: 100

57 Arrhenius & Catalysis Variant 57 — 9701/1/22/Q57

EASY · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 10
- B: 1
- C: 100
- D: 10000

58 Arrhenius & Catalysis Variant 58 — 9701/1/22/Q58

HARD · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 100
- B: 10
- C: 1
- D: 10000

59 Arrhenius & Catalysis Variant 59 — 9701/1/22/Q59

EASY · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 10
- B: 1
- C: 10000
- D: 100

60 Arrhenius & Catalysis Variant 60 — 9701/1/22/Q60

HARD · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 10
- B: 100
- C: 1
- D: 10000

61 Arrhenius & Catalysis Variant 61 — 9701/1/22/Q61

EASY · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 10
- B: 100
- C: 1
- D: 10000

62 Arrhenius & Catalysis Variant 62 — 9701/1/22/Q62

HARD · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 10
- B: 1
- C: 10000
- D: 100

63 Arrhenius & Catalysis Variant 63 — 9701/1/22/Q63

EASY · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 100
- B: 10
- C: 1
- D: 10000

64 Arrhenius & Catalysis Variant 64 — 9701/1/22/Q64

HARD · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 10
- B: 1
- C: 100
- D: 10000

65 Arrhenius & Catalysis Variant 65 — 9701/1/22/Q65

EASY · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 100
- B: 10
- C: 1
- D: 10000

66 Arrhenius & Catalysis Variant 66 — 9701/1/22/Q66

HARD · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 10
- B: 1
- C: 10000
- D: 100

67 Arrhenius & Catalysis Variant 67 — 9701/1/22/Q67

EASY · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 10
- B: 100
- C: 1
- D: 10000

68 Arrhenius & Catalysis Variant 68 — 9701/1/22/Q68

HARD · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 10
- B: 1
- C: 100
- D: 10000

69 Arrhenius & Catalysis Variant 69 — 9701/1/22/Q69

EASY · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 10
- B: 100
- C: 1
- D: 10000

70 Arrhenius & Catalysis Variant 70 — 9701/1/22/Q70

HARD · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 100
- B: 10
- C: 1
- D: 10000

71 Arrhenius & Catalysis Variant 71 — 9701/1/22/Q71

EASY · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 10
- B: 1
- C: 10000
- D: 100

72 Arrhenius & Catalysis Variant 72 — 9701/1/22/Q72

HARD · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 10
- B: 1
- C: 100
- D: 10000

73 Arrhenius & Catalysis Variant 73 — 9701/1/22/Q73

EASY · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 100
- B: 10
- C: 1
- D: 10000

74 Arrhenius & Catalysis Variant 74 — 9701/1/22/Q74

HARD · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 10
- B: 1
- C: 100
- D: 10000

75 Arrhenius & Catalysis Variant 75 — 9701/1/22/Q75

EASY · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 10
- B: 100
- C: 1
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76 Arrhenius & Catalysis Variant 76 — 9701/1/22/Q76

HARD · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 10
- B: 1
- C: 10000
- D: 100

77 Arrhenius & Catalysis Variant 77 — 9701/1/22/Q77

EASY · Syllabus 26.2

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- A: 10
- B: 1
- C: 100
- D: 10000

78 Arrhenius & Catalysis Variant 78 — 9701/1/22/Q78

HARD · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 100
- B: 10
- C: 1
- D: 10000

79 Arrhenius & Catalysis Variant 79 — 9701/1/22/Q79

EASY · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 10
- B: 1
- C: 10000
- D: 100

80 Arrhenius & Catalysis Variant 80 — 9701/1/22/Q80

HARD · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 10
- B: 100
- C: 1
- D: 10000

81 Arrhenius & Catalysis Variant 81 — 9701/1/22/Q81

EASY · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 10
- B: 100
- C: 1
- D: 10000

82 Arrhenius & Catalysis Variant 82 — 9701/1/22/Q82

HARD · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 10
- B: 1
- C: 10000
- D: 100

83 Arrhenius & Catalysis Variant 83 — 9701/1/22/Q83

EASY · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 100
- B: 10
- C: 1
- D: 10000

84 Arrhenius & Catalysis Variant 84 — 9701/1/22/Q84

HARD · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 10
- B: 1
- C: 100
- D: 10000

85 Arrhenius & Catalysis Variant 85 — 9701/1/22/Q85

EASY · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 100
- B: 10
- C: 1
- D: 10000

86 Arrhenius & Catalysis Variant 86 — 9701/1/22/Q86

HARD · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 10
- B: 1
- C: 10000
- D: 100

87 Arrhenius & Catalysis Variant 87 — 9701/1/22/Q87

EASY · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 10
- B: 100
- C: 1
- D: 10000

88 Arrhenius & Catalysis Variant 88 — 9701/1/22/Q88

HARD · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 10
- B: 1
- C: 100
- D: 10000

89 Arrhenius & Catalysis Variant 89 — 9701/1/22/Q89

EASY · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 10
- B: 100
- C: 1
- D: 10000

90 Arrhenius & Catalysis Variant 90 — 9701/1/22/Q90

HARD · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 100
- B: 10
- C: 1
- D: 10000

91 Arrhenius & Catalysis Variant 91 — 9701/1/22/Q91

EASY · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 10
- B: 1
- C: 10000
- D: 100

92 Arrhenius & Catalysis Variant 92 — 9701/1/22/Q92

HARD · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 10
- B: 1
- C: 100
- D: 10000

93 Arrhenius & Catalysis Variant 93 — 9701/1/22/Q93

EASY · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 100
- B: 10
- C: 1
- D: 10000

94 Arrhenius & Catalysis Variant 94 — 9701/1/22/Q94

HARD · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 10
- B: 1
- C: 100
- D: 10000

95 Arrhenius & Catalysis Variant 95 — 9701/1/22/Q95

EASY · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 10
- B: 100
- C: 1
- D: 10000

96 Arrhenius & Catalysis Variant 96 — 9701/1/22/Q96

HARD · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 10
- B: 1
- C: 10000
- D: 100

97 Arrhenius & Catalysis Variant 97 — 9701/1/22/Q97

EASY · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 10
- B: 1
- C: 100
- D: 10000

98 Arrhenius & Catalysis Variant 98 — 9701/1/22/Q98

HARD · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 100
- B: 10
- C: 1
- D: 10000

99 Arrhenius & Catalysis Variant 99 — 9701/1/22/Q99

EASY · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 10
- B: 1
- C: 10000
- D: 100

100 Arrhenius & Catalysis Variant 100 — 9701/1/22/Q100

HARD · Syllabus 26.2

A catalyst increases the forward rate of a reversible reaction by a factor of 100. By what factor is the rate of the reverse reaction increased?

- A: 10
- B: 100
- C: 1
- D: 10000

SUBTOPIC 26.1 — RATE EQUATIONS, ORDERS & HALF-LIFE (Q1 – Q50)**101 Core Repeat: Constant Half-Life Signature — 9701/11/M/J/23/Q13**

EASY · Syllabus 26.1

Which graph is the defining diagnostic test for a FIRST-ORDER reaction?

- A: A straight line on a concentration–time graph
- B: A curve of rate against concentration that passes through (0,0) parabolically
- C: A concentration–time graph showing constant successive half-lives
- D: A horizontal line on a rate–concentration graph

102 Core Repeat: Calculating k from Rate Law — 9701/12/M/J/23/Q14

HARD · Syllabus 26.1

For rate = $k[A][B]^2$, when $[A] = 0.10$ and $[B] = 0.20 \text{ mol dm}^{-3}$, rate = $8.0 \times 10^{-4} \text{ mol dm}^{-3} \text{ s}^{-1}$. What is k ?

- A: $0.20 \text{ mol}^{-2} \text{ dm}^6 \text{ s}^{-1}$
- B: $0.040 \text{ mol}^{-2} \text{ dm}^6 \text{ s}^{-1}$
- C: $0.40 \text{ mol}^{-2} \text{ dm}^6 \text{ s}^{-1}$
- D: $2.0 \text{ mol}^{-2} \text{ dm}^6 \text{ s}^{-1}$

103 Core Repeat: Order w.r.t. Catalyst — 9701/13/M/J/23/Q14

HARD · Syllabus 26.1

Can a catalyst appear in the rate equation for a reaction?

- A: No, catalysts are not consumed so they can never appear in rate equations.
- B: Only if the catalyst is a solid.
- C: Only if the reaction is endothermic.
- D: Yes, catalysts can appear in the rate equation with an order of reaction.

SUBTOPIC 26.2 — CATALYSIS & ARRHENIUS EQUATION (Q51 – Q100)

104 Core Repeat: Arrhenius Plot Gradient Formula — 9701/11/O/N/23/Q14

EASY · Syllabus 26.2

If the gradient of a plot of $\ln k$ vs $1/T$ is -12000 K , what is E_a ? ($R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$)

- A: $-99.8 \text{ kJ mol}^{-1}$
- B: $+99.8 \text{ kJ mol}^{-1}$
- C: $+1443 \text{ kJ mol}^{-1}$
- D: $+12.0 \text{ kJ mol}^{-1}$

SUBTOPIC 26.1 — RATE EQUATIONS, ORDERS & HALF-LIFE (Q1 – Q50)**105 Core Repeat: Zero-Order Rate-Concentration Plot — 9701/12/O/N/23/Q14**

EASY · Syllabus 26.1

What is the shape of a plot of rate against concentration for a zero-order reaction?

- A: A horizontal line parallel to the concentration axis
- B: A straight line of positive gradient passing through the origin
- C: An upward-curving parabola
- D: A vertical line parallel to the rate axis

106 Core Repeat: First-Order Half-Life Formula — 9701/13/O/N/23/Q14

EASY · Syllabus 26.1

Which equation correctly relates the half-life $t_{1/2}$ and rate constant k of a first-order reaction?

- A: $t_{1/2} = k / 0.693$
- B: $t_{1/2} = 0.693 \times k$
- C: $t_{1/2} = 0.693 / k$
- D: $t_{1/2} = 1 / (k \times [A]_0)$

SUBTOPIC 26.2 — CATALYSIS & ARRHENIUS EQUATION (Q51 – Q100)**107 Core Repeat: Poisoning of Catalytic Converters — 9701/11/M/J/22/Q14**

HARD · Syllabus 26.2

Why does lead in petrol poison the catalytic converter of a motor vehicle?

- A: Lead reacts exothermically with oxygen, melting the ceramic honeycomb.
- B: Lead atoms adsorb irreversibly onto the platinum/rhodium active sites, blocking reactant adsorption.
- C: Lead forms a gaseous complex with carbon monoxide.
- D: Lead increases the activation energy of the catalyst.

108 Core Repeat: Second-Order Half-Life Behavior — 9701/12/M/J/22/Q14

HARD · Syllabus 26.2

For a second-order reaction ($\text{rate} = k[A]^2$), what happens to each successive half-life as $[A]$ falls?

- A: Each successive half-life halves in duration.
- B: Each successive half-life remains strictly constant.
- C: The half-life drops to zero.
- D: Each successive half-life doubles in duration.

109 Core Repeat: Catalytic Converters Reagents and Products — 9701/13/M/J/22/Q14

EASY · Syllabus 26.2

Which reaction occurs on the surface of a three-way catalytic converter in a vehicle exhaust?

- A: $2\text{NO(g)} + 2\text{CO(g)} \rightarrow \text{N}_2\text{(g)} + 2\text{CO}_2\text{(g)}$
- B: $\text{N}_2\text{(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO(g)}$
- C: $\text{CO}_2\text{(g)} + \text{C(s)} \rightarrow 2\text{CO(g)}$
- D: $\text{SO}_2\text{(g)} + \text{O}_2\text{(g)} \rightarrow \text{SO}_3\text{(g)}$

110 Core Repeat: Autocatalysis Concept — 9701/11/O/N/22/Q14**HARD · Syllabus 26.2**

In the titration of ethanedioic acid with acidified potassium manganate(VII), the reaction starts slowly and then rapidly accelerates. Why?

- A: The temperature increases due to endothermic absorption.
- B: Potassium ions act as an enzyme.
- C: Mn^{2+} ions produced in the reaction act as an autocatalyst.
- D: Carbon dioxide gas dissolves to form an acid catalyst.

Quick-Check Answer Key Grid — Topic 26 — Reaction Kinetics (A Level MCQs)

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701) — OFFICIAL ANSWER KEY MATRIX

Q#	ANS	Q#	ANS	Q#	ANS	Q#	ANS	Q#	ANS
Q1	B	Q23	A	Q45	A	Q67	B	Q89	B
Q2	D	Q24	C	Q46	D	Q68	C	Q90	A
Q3	A	Q25	A	Q47	B	Q69	B	Q91	D
Q4	C	Q26	D	Q48	C	Q70	A	Q92	C
Q5	A	Q27	B	Q49	B	Q71	D	Q93	A
Q6	D	Q28	C	Q50	A	Q72	C	Q94	C
Q7	B	Q29	B	Q51	D	Q73	A	Q95	B
Q8	C	Q30	A	Q52	C	Q74	C	Q96	D
Q9	B	Q31	D	Q53	A	Q75	B	Q97	C
Q10	A	Q32	C	Q54	C	Q76	D	Q98	A
Q11	D	Q33	A	Q55	B	Q77	C	Q99	D
Q12	C	Q34	C	Q56	D	Q78	A	Q100	B
Q13	A	Q35	B	Q57	C	Q79	D	Q101	C
Q14	C	Q36	D	Q58	A	Q80	B	Q102	A
Q15	B	Q37	C	Q59	D	Q81	B	Q103	D
Q16	D	Q38	A	Q60	B	Q82	D	Q104	B
Q17	C	Q39	D	Q61	B	Q83	A	Q105	A
Q18	A	Q40	B	Q62	D	Q84	C	Q106	C
Q19	D	Q41	B	Q63	A	Q85	A	Q107	B
Q20	B	Q42	D	Q64	C	Q86	D	Q108	D
Q21	B	Q43	A	Q65	A	Q87	B	Q109	A
Q22	D	Q44	C	Q66	D	Q88	C	Q110	C

Explanations & Distractor Analysis — Topic 26 — Reaction Kinetics (A Level MCQs)

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701) — EXAMINER RATIONALE

1	B	Option B is correct. $\text{rate} = k[A]^2 \Rightarrow k = \text{rate} / [A]^2 = (\text{mol dm}^{-3} \text{ s}^{-1}) / (\text{mol dm}^{-3})^2 = \text{mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$.	[1]
2	D	Option D is correct. For a first-order reaction, $t_{1/2} = \ln 2 / k$. Because neither initial nor instantaneous concentration appears in this expression, the half-life is strictly constant throughout the entire course of the reaction.	[1]
3	A	Option A is correct. Rate $[A]^m: 2^m = 4 \Rightarrow m = 2$ (second order in A). Rate $[B]^n: 2^n = 1 \Rightarrow n = 0$ (zero order in B). Thus $\text{rate} = k[A]^2[B]^0 = k[A]^2$.	[1]
4	C	Option C is correct. The exponents in the experimental rate equation correspond directly to the stoichiometric molecularity of the reactants in the rate-determining (slowest) elementary step.	[1]
5	A	Option A is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0 \text{ s}$.	[1]
6	D	Option D is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0 \text{ s}$.	[1]
7	B	Option B is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0 \text{ s}$.	[1]
8	C	Option C is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0 \text{ s}$.	[1]
9	B	Option B is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0 \text{ s}$.	[1]
10	A	Option A is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0 \text{ s}$.	[1]
11	D	Option D is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0 \text{ s}$.	[1]
12	C	Option C is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0 \text{ s}$.	[1]
13	A	Option A is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0 \text{ s}$.	[1]
14	C	Option C is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0 \text{ s}$.	[1]
15	B	Option B is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0 \text{ s}$.	[1]
16	D	Option D is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0 \text{ s}$.	[1]
17	C	Option C is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0 \text{ s}$.	[1]
18	A	Option A is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0 \text{ s}$.	[1]
19	D	Option D is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0 \text{ s}$.	[1]
20	B	Option B is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0 \text{ s}$.	[1]
21	B	Option B is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0 \text{ s}$.	[1]
22	D	Option D is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0 \text{ s}$.	[1]
23	A	Option A is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0 \text{ s}$.	[1]
24	C	Option C is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0 \text{ s}$.	[1]
25	A	Option A is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0 \text{ s}$.	[1]
26	D	Option D is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0 \text{ s}$.	[1]
27	B	Option B is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0 \text{ s}$.	[1]
28	C	Option C is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0 \text{ s}$.	[1]
29	B	Option B is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0 \text{ s}$.	[1]
30	A	Option A is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0 \text{ s}$.	[1]
31	D	Option D is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0 \text{ s}$.	[1]

32	C	Option C is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0$ s.	[1]
33	A	Option A is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0$ s.	[1]
34	C	Option C is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0$ s.	[1]
35	B	Option B is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0$ s.	[1]
36	D	Option D is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0$ s.	[1]
37	C	Option C is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0$ s.	[1]
38	A	Option A is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0$ s.	[1]
39	D	Option D is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0$ s.	[1]
40	B	Option B is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0$ s.	[1]
41	B	Option B is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0$ s.	[1]
42	D	Option D is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0$ s.	[1]
43	A	Option A is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0$ s.	[1]
44	C	Option C is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0$ s.	[1]
45	A	Option A is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0$ s.	[1]
46	D	Option D is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0$ s.	[1]
47	B	Option B is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0$ s.	[1]
48	C	Option C is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0$ s.	[1]
49	B	Option B is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0$ s.	[1]
50	A	Option A is correct. $t_{1/2} = 0.693 / k = 0.693 / 0.0693 = 10.0$ s.	[1]
51	D	Option D is correct. Comparing $\ln k = -(E_a/R)(1/T) + \ln A$ with $y = mx + c$ reveals that $y = \ln k$, $x = 1/T$, slope $m = -E_a/R$, and y-intercept $c = \ln A$.	[1]
52	C	Option C is correct. Fe^{2+} reduces $\text{S}_2\text{O}_8^{2-}$ to SO_4^{2-} while being oxidised to Fe^{3+} , and Fe^{3+} subsequently oxidises I^- to I_2 while being regenerated back to Fe^{2+} . Both steps involve attractive forces between cation and anion, bypassing the high electrostatic repulsion of the uncatalysed anion-anion reaction.	[1]
53	A	Option A is correct. Adsorption involves formation of temporary bonds between reactant molecules and active surface sites. This strains and weakens internal reactant bonds and concentrates the molecules in favorable orientations, lowering E_a .	[1]
54	C	Option C is correct. Collision frequency increases by only ~2% for a 10 K rise. The dramatic rate acceleration is due to the exponential increase in the fraction of particles possessing energy equal to or greater than the activation energy ($e^{(-E_a/RT)}$).	[1]
55	B	Option B is correct. A catalyst lowers the activation energy of both the forward and reverse reactions by the exact same amount. Consequently, both forward and reverse rates increase by the identical factor, leaving the position of equilibrium (and K_c) unchanged.	[1]
56	D	Option D is correct. A catalyst lowers the activation energy of both the forward and reverse reactions by the exact same amount. Consequently, both forward and reverse rates increase by the identical factor, leaving the position of equilibrium (and K_c) unchanged.	[1]
57	C	Option C is correct. A catalyst lowers the activation energy of both the forward and reverse reactions by the exact same amount. Consequently, both forward and reverse rates increase by the identical factor, leaving the position of equilibrium (and K_c) unchanged.	[1]
58	A	Option A is correct. A catalyst lowers the activation energy of both the forward and reverse reactions by the exact same amount. Consequently, both forward and reverse rates increase by the identical factor, leaving the position of equilibrium (and K_c) unchanged.	[1]
59	D	Option D is correct. A catalyst lowers the activation energy of both the forward and reverse reactions by the exact same amount. Consequently, both forward and reverse rates increase by the identical factor, leaving the position of equilibrium (and K_c) unchanged.	[1]
60	B	Option B is correct. A catalyst lowers the activation energy of both the forward and reverse reactions by the exact same amount. Consequently, both forward and reverse rates increase by the identical factor, leaving the position of equilibrium (and K_c) unchanged.	[1]
61	B	Option B is correct. A catalyst lowers the activation energy of both the forward and reverse reactions by the exact same amount. Consequently, both forward and reverse rates increase by the identical factor, leaving the position of equilibrium (and K_c) unchanged.	[1]

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89	B	Option B is correct. A catalyst lowers the activation energy of both the forward and reverse reactions by the exact same amount. Consequently, both forward and reverse rates increase by the identical factor, leaving the position of equilibrium (and K_c) unchanged. [1]
90	A	Option A is correct. A catalyst lowers the activation energy of both the forward and reverse reactions by the exact same amount. Consequently, both forward and reverse rates increase by the identical factor, leaving the position of equilibrium (and K_c) unchanged. [1]
91	D	Option D is correct. A catalyst lowers the activation energy of both the forward and reverse reactions by the exact same amount. Consequently, both forward and reverse rates increase by the identical factor, leaving the position of equilibrium (and K_c) unchanged. [1]
92	C	Option C is correct. A catalyst lowers the activation energy of both the forward and reverse reactions by the exact same amount. Consequently, both forward and reverse rates increase by the identical factor, leaving the position of equilibrium (and K_c) unchanged. [1]
93	A	Option A is correct. A catalyst lowers the activation energy of both the forward and reverse reactions by the exact same amount. Consequently, both forward and reverse rates increase by the identical factor, leaving the position of equilibrium (and K_c) unchanged. [1]
94	C	Option C is correct. A catalyst lowers the activation energy of both the forward and reverse reactions by the exact same amount. Consequently, both forward and reverse rates increase by the identical factor, leaving the position of equilibrium (and K_c) unchanged. [1]
95	B	Option B is correct. A catalyst lowers the activation energy of both the forward and reverse reactions by the exact same amount. Consequently, both forward and reverse rates increase by the identical factor, leaving the position of equilibrium (and K_c) unchanged. [1]
96	D	Option D is correct. A catalyst lowers the activation energy of both the forward and reverse reactions by the exact same amount. Consequently, both forward and reverse rates increase by the identical factor, leaving the position of equilibrium (and K_c) unchanged. [1]
97	C	Option C is correct. A catalyst lowers the activation energy of both the forward and reverse reactions by the exact same amount. Consequently, both forward and reverse rates increase by the identical factor, leaving the position of equilibrium (and K_c) unchanged. [1]
98	A	Option A is correct. A catalyst lowers the activation energy of both the forward and reverse reactions by the exact same amount. Consequently, both forward and reverse rates increase by the identical factor, leaving the position of equilibrium (and K_c) unchanged. [1]
99	D	Option D is correct. A catalyst lowers the activation energy of both the forward and reverse reactions by the exact same amount. Consequently, both forward and reverse rates increase by the identical factor, leaving the position of equilibrium (and K_c) unchanged. [1]
100	B	Option B is correct. A catalyst lowers the activation energy of both the forward and reverse reactions by the exact same amount. Consequently, both forward and reverse rates increase by the identical factor, leaving the position of equilibrium (and K_c) unchanged. [1]
101	C	Option C is correct. Constant half-life on a concentration-time plot is the definitive experimental hallmark of first-order kinetics. [1]
102	A	Option A is correct. $k = \text{rate} / ([A][B]^2) = 8.0 \times 10^{-4} / (0.10 \times 0.040) = 8.0 \times 10^{-4} / 4.0 \times 10^{-3} = 0.20 \text{ mol}^{-2} \text{ dm}^6 \text{ s}^{-1}$. [1]
103	D	Option D is correct. Homogeneous catalysts participate directly in the rate-determining step and their concentration directly influences the reaction rate, giving them a non-zero order in the rate equation (e.g. $\text{rate} = k [\text{propanone}][\text{H}^+]$). [1]
104	B	Option B is correct. $\text{Gradient} = -E_a / R \Rightarrow E_a = -(-12000 \text{ K}) \times 8.314 \text{ J K}^{-1} \text{ mol}^{-1} = +99768 \text{ J mol}^{-1} = +99.8 \text{ kJ mol}^{-1}$. [1]
105	A	Option A is correct. Zero order means $\text{rate} = k[A]^0 = k$ (constant). The rate does not change as concentration changes, yielding a horizontal line. [1]
106	C	Option C is correct. $t_{1/2} = \ln 2 / k \approx 0.693 / k$. [1]
107	B	Option B is correct. Heavy metals like lead bind strongly and irreversibly to surface metal sites (permanent chemisorption), preventing reactant molecules from adsorbing. [1]
108	D	Option D is correct. For a second-order reaction, $t_{1/2} = 1 / (k[A]_0)$. Halving the concentration doubles the subsequent half-life. [1]
109	A	Option A is correct. The converter catalyzes the redox reaction between pollutant gases NO and CO to yield harmless N_2 and CO_2 . [1]
110	C	Option C is correct. This is classic autocatalysis: the product $\text{Mn}^{2+}(\text{aq})$ acts as a homogeneous catalyst, accelerating the reaction as it accumulates. [1]